Document for

Generalized Open-set Single-shot Character Recognition on Ancient Egyptian Hieratic Characters

Requirement / Test bench spec

CPU: anything with AVX2 / R5 5500

* Intel or AMD, I have no idea if this works on Zhaoxin or smth to that nature

GPU: Nvidia GPU (Pascal or newer) / GTX1070

Ram: 8 Gib/ 32Gib

Disk: Yes / 320Gib HDD

OS: linux / Debian 12.14

Internet: Yes / Yes

Note: this document is not for power users.

This is NOT how things should be done,

but rather a quick way to just run it on a testbench.

Audit your code and script and set environments properly,

and please do not blindly run random bash scripts off internet.

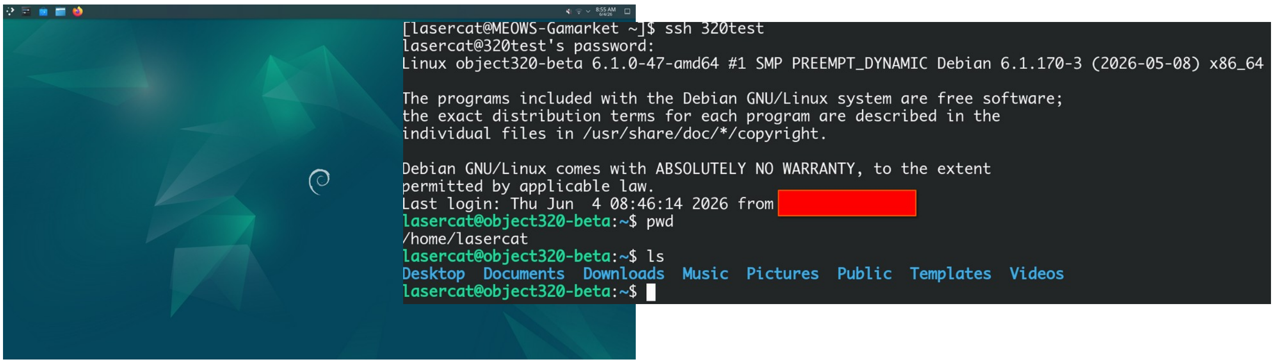
Trivia: Maunl, aka Pallas’ cat, is the oldest cat specie on earth today.

1. Setup Environment

System &Cuda

1. install debian 12.14 KDE, select ssh if you need it, use lasercat as username

you can mod the code later to fit your username and paths...



2. download script from https://github.com/lancercat/320-env-debian to ~/Downloads

3. set up basic environment

```
su
```

```
bash basicenv.sh
```

4. **reboot** (bcs the kernel version may bump, so this is a **MUST**)

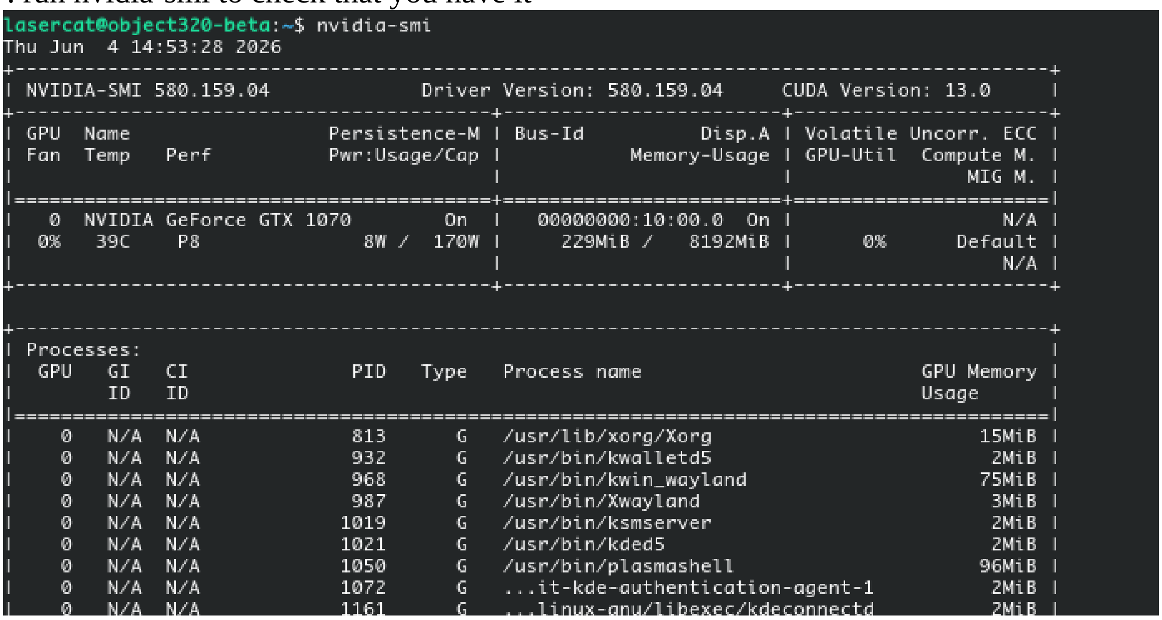
5. install cuda

```
su
```

```
bash setupenv.sh
```

6. reboot (for nvidia driver to kick in)

7. run nvidia-smi to check that you have it



Torch & Etc

1. install torch and torch_scatter (a custom fork to fix glog)

```
source ~/.bashrc && bash pyenv.sh
```

2. set up code, weight, and models.

a. prepare dir:

sh makdirs.sh

b. clone code

cd ~/cat; git clone git@github.com:lancercat/object-320-beta.git

c. download data and weight to Downloads, put weights to ~/hydra_saves and data to ~/ssddata

3. check that you have everything

```
lasercat@object320-beta:~$ ls ~/cat
object-320-beta
lasercat@object320-beta:~$ ls ~/ssddata/
dicts_v3  ecr-v3R
lasercat@object320-beta:~$ ls ~/hydra_saves/
b1nf-1024-ecrv3R-abiaug-48-48       b1nf-1024-ecrv3R-noaug-48-48       b1nf-1024-ecrv3RTW-abiaug-48-48
b1nf-1024-ecrv3R-abiaug-48-48-run2  b1nf-1024-ecrv3R-noaug-48-48-run2  b1nf-1024-ecrv3RTW-abiaug-48-48-run2
```

2. Reproducing the results

1. limit your card if you have a shoddy psu

su

nvidia-smi -pm 1 && nvidia-smi -pl 100

2. active your torch

```
source ~/Downloads/env/pytorch_env/bin/activate
```

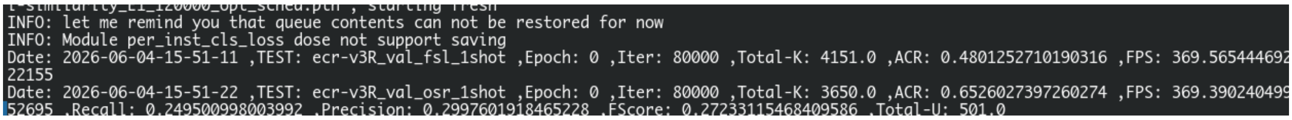
3. go to eval script dir

```
cd ~/cat/object-320-beta/neko_2025_NGNW/beta320
```

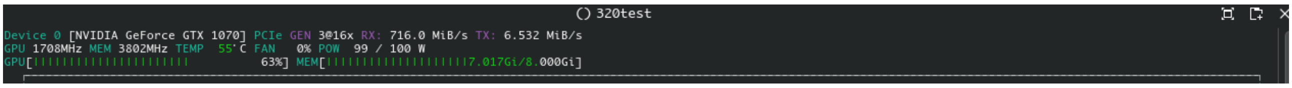
4. run inference

```
PYTHONPATH=${PWD}/../../../ python eval_batch.py | tee hee.log
```

if you see this, its running,



utilization btw, note we are running off hdd so the io is not ideal.

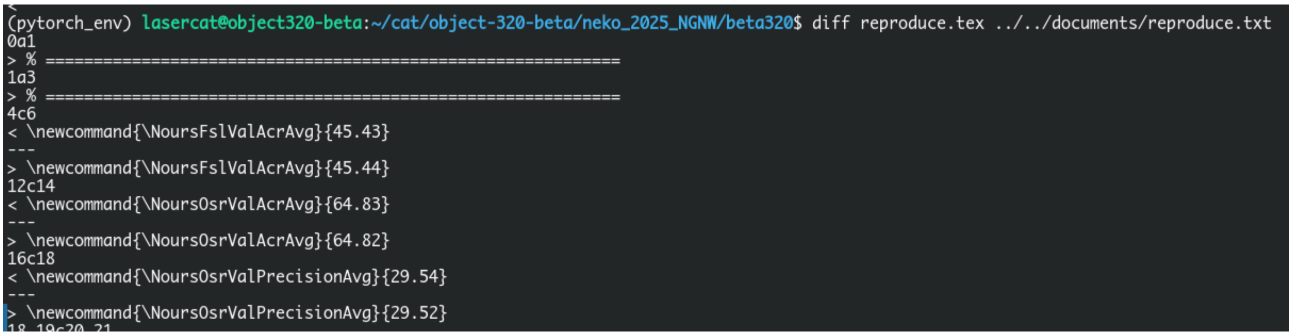


5. extrat performance

```
awk '/% ====/ {flag=1; next} flag' hee.log > reproduce.
tex
```

5. compare it to the paper report value

Note there could be minor performance differences moving from environment to environment



3. Train a model

1. limit your card if your psu came from a pawn shop or smth

su

nvidia-smi -pm 1 && nvidia-smi -pl 100

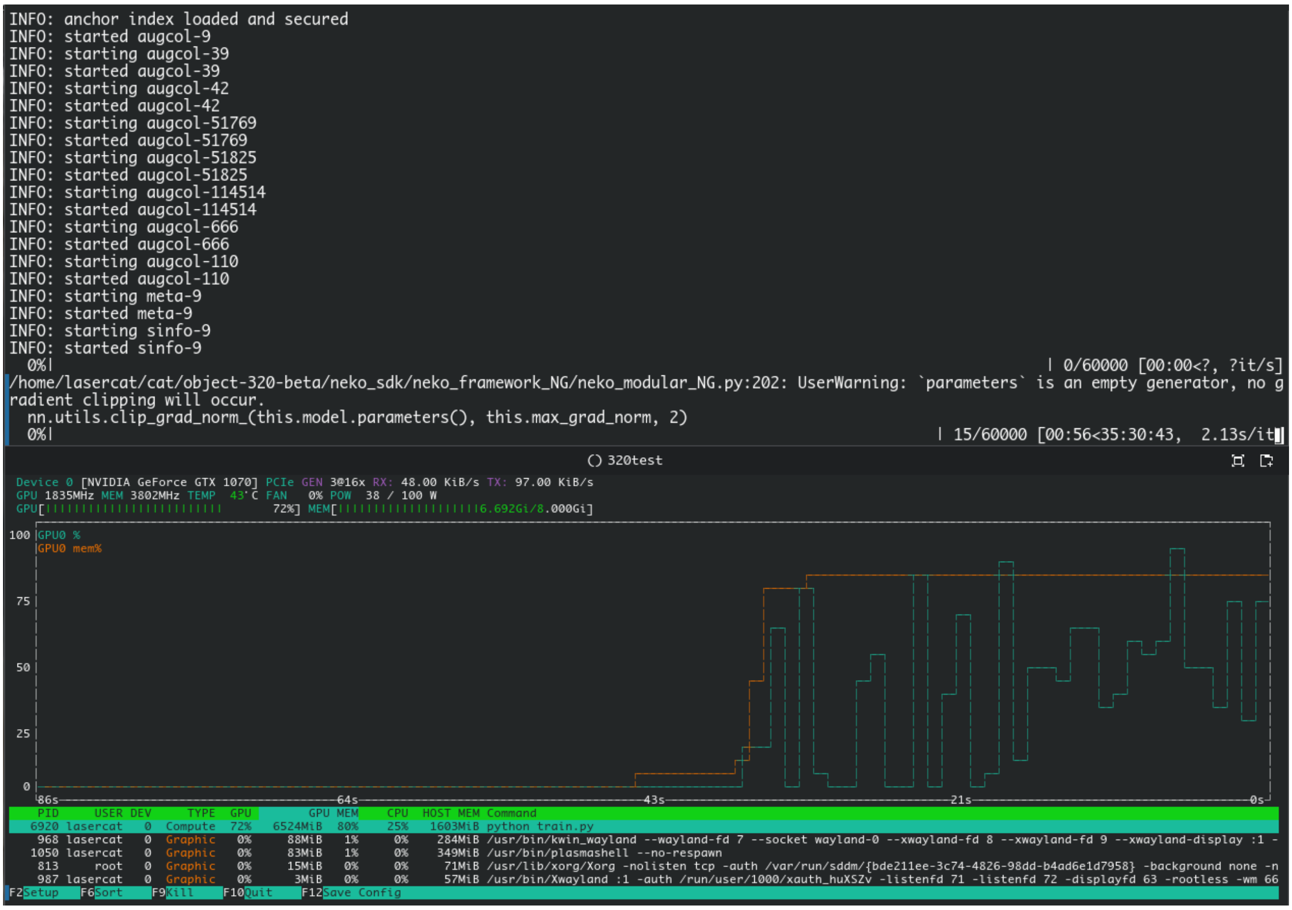
2. go to a model dir, say b1nf-1024-ecrv3R-noaug-48-48

`cd ~/cat/object-320-beta/neko_2025_NGNW/beta320/b1nf-1024-ecrv3R-noaug-48-48`

3. train

`PYTHONPATH=${PWD}/../../../ python train.py | tee hee.log`

If you see this, you are training. Utilization btw



That's all for now, GLHF